Linear Regression

1. Which of the following best describes the goal of linear regression?
   A. Classify data into categories
   B. Find clusters in data
   C. Model relationship between dependent and independent variables
   D. Reduce dimensionality of data

2. In linear regression, the relationship between variables is assumed to be:
   A. Always exponential
   B. Always linear in parameters
   C. Always nonlinear
   D. Random

3. What does the error term in linear regression represent?
   A. Perfect prediction
   B. Unexplained variation in data
   C. Feature scaling factor
   D. Model accuracy

4. Which metric is most sensitive to outliers?
   A. MAE
   B. MSE
   C. R^2
   D. Accuracy

5. If $R^2 = 0$, it means:
   A. Perfect model
   B. No relationship explained by model
   C. Model is overfitting
   D. Model is correct

6. Multicollinearity affects:
   A. Only target variable
   B. Relationship between independent variables
   C. Only categorical variables
   D. Only clustering models

7. Which assumption is NOT required in linear regression?
   A. Linearity
   B. Independence
   C. Homoscedasticity
   D. Normal distribution of input features

8. Gradient descent is used to:
   A. Reduce model error
   B. Increase data size
   C. Normalize features
   D. Remove features

9. Which transformation helps handle skewed data?
   A. One-hot encoding
   B. Log transformation
   C. PCA
   D. Label encoding

10. In simple linear regression, how many independent variables are used?
    A. 0
    B. 1
    C. 2
    D. Many

11. If adding more features always increases R^2, what risk arises?
    A. Underfitting
    B**. Overfitting**
    C. Data leakage
    D. Feature loss

12. A high training R^2 but low test R^2 indicates:
    A. Underfitting
    B. Good generalization
    C. Overfitting
    D. No learning

13. Which situation violates linear regression assumptions most?
    A. Linear relationship
    B. Constant variance
    C. Non-linear relationship between features and target
    D. Independent samples

14. If residuals show a pattern, it suggests:
    A. Model is perfect
    B. Model is biased
    C. Model is linear
    D. Data is clean

15. Which feature type requires encoding before regression?
    A. Continuous
    B. Integer
    C. Categorical
    D. Scaled numeric

16. A model with high bias usually:
    A. Overfits
    B. Underfits
    C. Memorizes data
    D. Has high variance

17. Increasing training data usually reduces:
    A. Bias
    B. Variance
    C. Features
    D. R^2

18. What happens if two features are highly correlated?
    A. Better model stability
    B. Multicollinearity issue
    C. Lower error automatically
    D. Feature scaling improves

19. Which is NOT a regression metric?
    A. RMSE
    B. MAE

C. Precision
D. R^2

20. The best-fit line minimizes:
    A. Number of points
    B**. Sum of squared errors**
    C. Number of features
    D. Mean of data

21. What does this code do?

    "model.fit(X_train, y_train)"

    A. Predicts values
    B. Trains model
    C. Evaluates model
    D. Splits dataset

22. What is y_pred in regression?

    "y_pred = model.predict(X_test)"

    A. True labels
    B. Predicted outputs
    C. Training inputs
    D. Error values

23. What does this function compute?

    "mean_squared_error(y_test, y_pred)"

    A. Accuracy
    B. Classification score
    C. Average squared error
    D. Correlation

24. What will happen if you forget scaling before regression?
    A. Always improves accuracy
    B. No effect
    C. Can distort model performance
    D. Converts regression to classification

25. What does train_test_split ensure?
    A. Data duplication
    B. Random separation of data
    C. Feature reduction
    D. Label encoding

26. In this code, what is missing?

    "model = LinearRegression()"
    "model.fit(X, y)"

    A. Prediction step
    B. Feature scaling
    C. Data splitting
    D. All of the above

27. What does model.coef_ represent?
    A. Intercept

B. Feature weights
C. Prediction error
D. Dataset size

28. What does this line compute?

    "model.intercept_"

    A. Feature importance
    B. Bias term
    C. Loss function
    D. Gradient

29. Which is correct about .predict()?
    A. Used for training
    B. Used for evaluation only
    C. Generates output predictions
    D. Splits data

30. What happens if y is categorical in linear regression?
    A. Works normally
    B. Becomes classification problem
    C. Model fails or gives invalid results
    D. Improves accuracy

31. What is the output shape of predict(X) typically?
    A. Same as X
    B. Same as y
    C. Scalar
    D. Random

32. Which library is commonly used?
    A. seaborn
    B. sklearn
    C. tensorflow only
    D. nltk

33. What does this code compute? *"r2_score(y_test, y_pred)"*

    A. Error magnitude
    B. Model fit quality
    C. Classification accuracy
    D. Clustering score

34. What happens if you train on full dataset without test split?
    A. Better generalization
    B. Overfitting cannot be detected
    C. Faster training
    D. Improved bias

35. What does normalization affect?
    A. Model architecture
    B. Feature scale
    C. Target variable only
    D. Dataset size

36. If house price prediction model fails in real world despite high training accuracy, likely issue is:

A. Underfitting
B. Overfitting
C. Data cleaning
D. Feature encoding

37. A dataset with strong seasonal pattern needs:
A. No preprocessing
B. Feature engineering
C. Removing target
D. Random sampling only

38. If residual plot shows funnel shape, it indicates:
A. Constant variance
B. Heteroscedasticity
C. Linear relationship
D. No error

39. Adding irrelevant features usually:
A. Always improves model
B. Increases noise
C. Removes bias
D. Reduces dataset size

40. Which model is most interpretable?
A. Neural network
B. Random forest
C. Linear regression
D. SVM

41. If two features are identical, regression will:
A. Improve accuracy
B. Fail or become unstable
C. Ignore both
D. Convert to classification

42. Outliers in target variable mostly affect:
A. Classification only
B. Regression strongly
C. Clustering only
D. PCA only

43. If slope is zero in regression line, it means:
A. Perfect prediction
B. No relationship between variables
C. Strong correlation
D. Overfitting

44. Increasing polynomial degree in regression:
A. Reduces model complexity
B. Increases non-linearity
C. Removes features
D. Improves interpretability

45. If all predictions are same value, model is:
A. Well trained
B**. Underfitting**

C. Overfitting
D. Perfect

46. If dataset size increases significantly, variance usually:
A. Increases
B. Decreases
C. Stays same
D. Becomes zero

47. Which is most important for regression performance?
A. Number of rows
B. Feature quality
C. File format
D. Color of data

48. A model with low training error but high test error is:
A. Generalized
B. Overfitted
C. Underfitted
D. Optimal

49. If relationship between X and y is exponential, linear regression:
A. Always perfect
B. May perform poorly
C. Always fails
D. Becomes classification

50. Best way to improve linear regression performance is usually:
A. Increase noise
B. Better feature engineering
C. Remove training data
D. Reduce test set